

The Economy

Seminar 5: Firms, Consumers and Markets

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Courtesy of
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The Firm

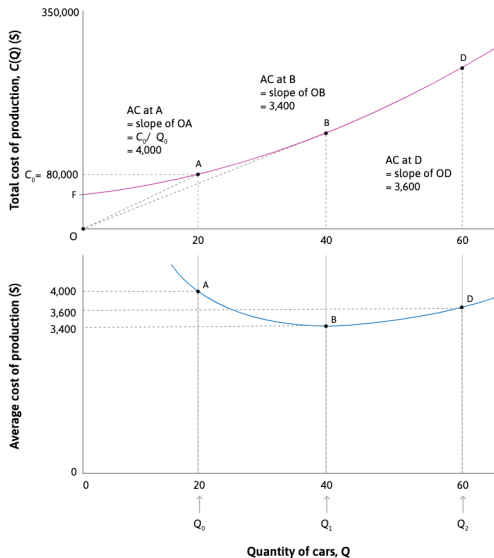
- we want to coordinate the actions (choices) of agents across an economy
 - we construct a **firm**: agents interact via **contracts** that establish the **hierarchy** of authority
 - we construct a **market**: agents interact via **prices** and power is "endogenously" spread through the market (no clear hierarchy, no central power)
- a **firm** employs individuals (L) and invests in physical/intangible capital (K) and uses them to convert inputs (production factors) into output (Y , Q)
- when firms choose in-house production over outsourcing (going to the market) if it **costs too much to produce in-house** or if there are **intricacies** (uncertainties, moral hazards) they want to **control**
 - firms would only outsource to the market activities for which the deliverable end result is unambiguously specifiable in the call for tender/subcontracting (if not, why give up the control?)

Firms: From Inputs to Outputs

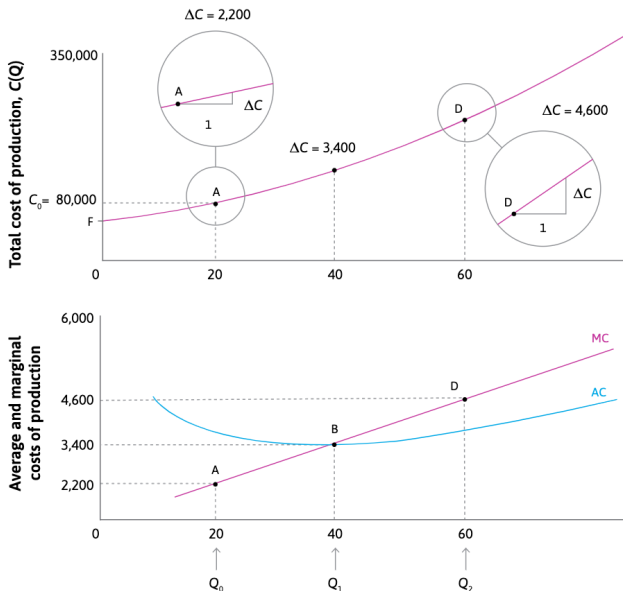
- to maximise their profits, firms take inputs to produce outputs and incur **costs of production** doing so
 - define a **cost function**, $C(Q)$, which tells us the cost needed to produce a given quantity Q of product
 - a cost of production can be **fixed**, denoted F , in that it is independent of the output (machines, rent) or **variable**, $C_V(Q)$, that is dependent on the level of output produced (wages, input costs, etc.)
 - so total = fixed + variable, i.e $C(Q) = F + C_V(Q)$
 - assume: that the variable cost function $C_V(Q)$ and the total cost function $C(Q)$ are convex
- define a **marginal cost**, denoted MC or $C_m(Q)$ as the cost of producing an extra unit of a good
 - this is the **slope of the cost function**, and thus changes with the amount of good produced
 - typically convex cost functions mean marginal costs is higher as the quantity of goods produced increases
- define the **average cost**, denoted AC or $C_M(Q)$ as the cost per unit produced
 - = the total cost divided by the quantity of goods produced, $C_M(Q) = \frac{C(Q)}{Q}$

- Notice that $C_M(Q) = \underbrace{\frac{F}{Q}}_{\text{decreasing in } Q} + \underbrace{\frac{C_V(Q)}{Q}}_{\text{increasing in } Q}$
 - the variable cost $C_V(Q)$ increases as Q increases, but by increasing larger marginal amounts ($C_V(Q)$ is convex)
- so at some quantity $\bar{Q}$, the rising variable cost must exactly balance out the decreasing fixed cost:
 - at $Q < \bar{Q}$, the amortisation of fixed cost dominates ($C_M(Q)$ is decreasing) + each additional unit is less costly than the average cost of the previous ones, so $C_M(Q) > C_m(Q)$
(i.e. at low Q , the marginal cost is low while the average cost is high)
 - at $Q > \bar{Q}$, the rising variable cost dominates ($C_M(Q)$ is increasing) + each additional unit is more costly than the average cost of the previous ones, so $C_M(Q) < C_m(Q)$
(i.e. at high Q , the marginal cost is very high while the average cost is now lower)
 - at $Q = \bar{Q}$, the average cost is at a minimum (average fixed cost is low enough and average variable cost is not too high) and any additional unit produced costs the average cost: $C_M(Q) = C_m(Q)$
(i.e. at this point, the MC has no effect on the AC: it neither drags it up nor pulls it down, and this can only be if the MC adds exactly the same cost as the AC originally was)

Costs



Costs



Profit

- firms maximise their profit = total revenue - total cost
 - the total revenue is the amount of output they sell $\times$ the price they sell the output for: $TR = P \times Q$
 - the total cost is $C(Q)$, as before
- then profit is $\Pi = TR - TC = PQ - C(Q)$
 - recall that the average cost $C_M(Q) = C(Q)/Q$, then the total cost can be written as $C(Q) = C_M(Q) \times Q$
 - then write profit as $\Pi = P \times Q - C_M(Q) \times Q = Q(P - C_M(Q))$
 - profit is positive if $P > C_M(Q)$ (price is larger than the average cost to produce)

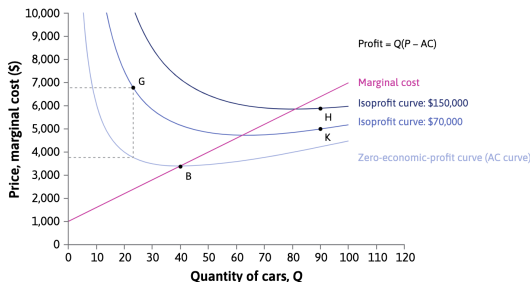
$$(P - C_M(Q)) \times Q = \Pi$$

$$P - C_M(Q) = \frac{\Pi}{Q}$$

$$\underbrace{P}_{\text{price}} = \underbrace{C_M(Q)}_{\text{AC per unit}} + \underbrace{\frac{\Pi}{Q}}_{\text{profit per unit}}$$

this gives an expression of P in terms of Q which links all combinations of (P, Q) to a given level of profit Π = isoprofit curve

Isoprofit Curves



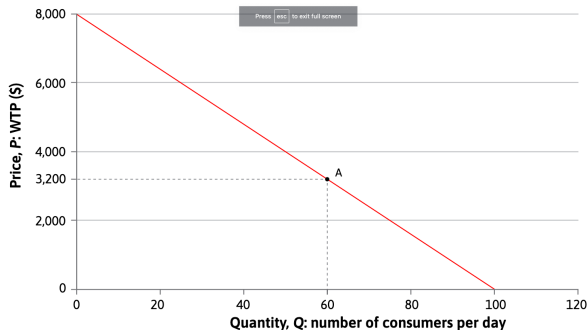
- isoprofit curves represent a given profit level: the further from the **horizontal axis**, the higher the profit (NB: not the origin)
 - $\Rightarrow$ in fact since $P = AC + \text{profit/unit}$, we have that each isoprofit curve are just upward shifts of each other, with profit per unit the size of the shift from the zero-profit curve (AC curve)
- the slope represents the MRS between preferring a high price to preferring a high quantity in making the same profit
 - *what is the change in price needed in order to sell one extra unit of quantity? OR what is the change in quantity needed in order to change price by one unit?*

Isoprofit Curves

$$\text{slope of isoprofit curve} = -\frac{P - MC}{Q} = -\frac{\text{profit margin}}{\text{quantity}}$$

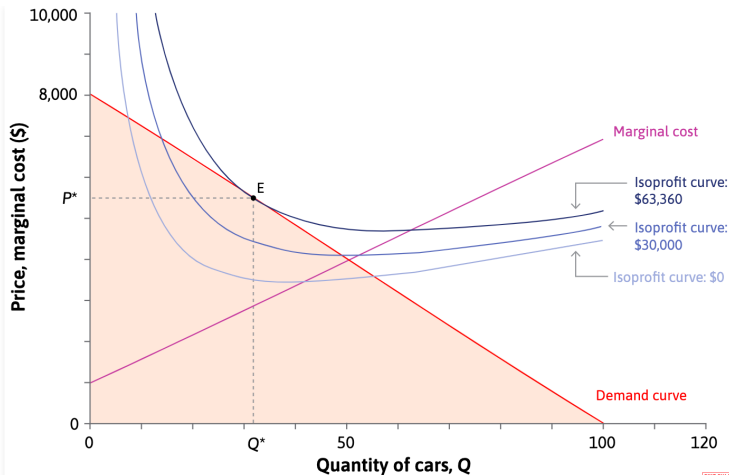
- ⇒ increasing quantity by 1 reduces P by the slope (so by $(P - MC)/Q$)
- ⇒ intuitively, if we produce one more unit but a lower price, then increase in revenue from the extra unit ($P - MC$) is then divided equally among all the other units (divide by Q) so the profit stays the same
- all then depends on the profit margin:
 - if $P > MC$, then the profit margin $P - MC$ is positive so the slope is negative: lower prices in order to sell an extra unit of quantity to keep profit constant
⇒ the isoprofit curve is downwards sloping
 - if $P < MC$, then the profit margin $P - MC$ is negative and the slope is positive: raise prices in order to sell an extra unit of quantity to keep profit constant
⇒ the isoprofit curve is upwards sloping
 - if $P = MC$, then the profit margin is zero, the slope is zero and we are at the minimum price (selling more means raising the price to keep costs down; selling less also means raising the price) ⇒ the consequence of which: **isoprofit curves intersect with the marginal cost curve at their minimum**
 - so at zero profits, price = average cost (and so no surprise that the MC curve also crosses this zero-profit curve at the minimum)

Demand Curve



- for a given price, how many units will be bought $\Rightarrow$ downward sloping since goods are usually ordinary
- a consumer with a willingness to pay (WTP) higher than the price will buy: so as price falls, demand increases
- think of the demand curve as the production possibility frontier from price and quantity: so its slope is the MRT of quantity into price (to sell an extra unit of quantity, how much lower of a price does a firm need to choose)

Profit Maximization



EXIT FULLSCREEN

Profit Maximisation and Price-Setting

- at the optimum E : choose a (price, quantity)-pair such that $MRS = MRT$
 - this is when for an extra unit of quantity sold, the amount that the price would need to come down by both (1) allows more consumers to buy (MRT) and (2) while keeping the profit level constant (MRS)
 - if $MRS > MRT$ (to the left of the optimum), then increasing the quantity sold by one unit would lower the price for the consumers too little to keep the firm on the same profit level (the profit is too low), so we can increase the profit by selling even more for a lower price
 - if $MRS < MRT$ (to the right of the optimum), then increasing the quantity sold by one unit would lower the price for the consumers too much to keep the firm on the same profit level, so we can increase the profit by selling less for a higher price
 - at the optimum, it is impossible to increase the profit by changing prices
 - additionally, since $MRS = MRT$, it must be that the slope of the isoprofit curve = slope of the demand curve (always downwards sloping), so the equilibrium is always such that $P > MC$

Profit Maximisation vis MR and MC

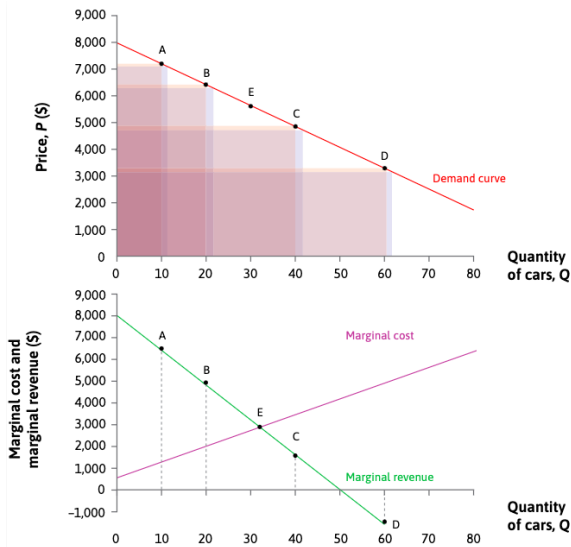
- so far we have defined everything in terms of prices and quantities, but we can also do so in terms of marginal revenue and marginal cost
 - **marginal revenue**, R_m , is the extra revenue made from selling an additional unit (= the demand curve)
 - recall $R = P \times Q$, then the marginal revenue is given by

$$R_m = \frac{\Delta R}{\Delta Q} = \frac{R' - R}{Q' - Q} = \frac{P'Q' - PQ}{1}$$

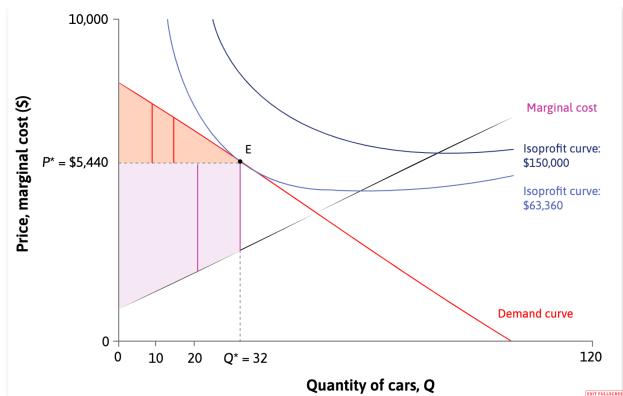
where P and Q are the current price and quantities and P' and Q' are the new prices and quantities once a marginal unit is produced and sold

- alternatively you can see this as the gain in producing and selling an extra unit - the loss in price on all previously sold units ($P' - Q(P' - P)$)
 - this is the production equivalent of the demand function (demanding one extra unit is costly but lowers the price of all the units demanded)
- **marginal cost**, C_m , is the cost of producing an extra unit (as before)
- profit is maximised when $R_m = C_m$ (this can be derived mathematically as the FOC of the profit maximisation problem)
 - if $R_m > C_m$, then we should produce and sell one more unit as there is still profit to be made on the next unit (marginal profit > 0)
 - if $R_m < C_m$, then we should produce and sell one less unit as we are currently making a loss on this unit (marginal profit < 0)

Profit Maximisation: $R_M = C_M$



Gains of Trade



- notice that when a good is traded, it is because both consumers and firms want to: **benefits of trade**
 - a firm benefits because it makes a profit $\Pi = P \times Q - C(Q) > 0$ (since $P > C_m$ or $R_m > C_m$) = **firm surplus**
 - a consumer benefits because his willingness to pay is higher than what he actually had to pay (= it's like he saved) - **consumer surplus**

Losses in Trade

- the "unequal" distribution of firm surplus and consumer surplus is because firms set a price above the marginal cost
 - if price = marginal cost, recall that this is at the minimum of the isoprofit curve, so the slope of the isoprofit curve is zero
 - but the slope of the demand curve is never zero
 - so they can never be an optimal point where a single firm would set its price = marginal cost
- but the point where price = marginal cost is where the total surplus is maximised and so setting a price $>$ marginal cost is Pareto inefficient
 - setting the price $>$ marginal cost excludes any consumer whose WTP is between marginal cost and current price, all of whom could have generated consumer surplus
 - so of this consumer surplus is converted into producer surplus, but the rest is **deadweight loss** = potential surplus that is never enjoyed - so this is Pareto inefficient but deeply in the producer's favour
 - to try and make this Pareto efficient we would need to force the firm to produce at price = marginal cost BUT also make sure the firm is still as well off as before by transferring some of the newly-gained consumer surplus to the firm (there is no guarantee this is possible)
- how feasible a Pareto improvement is depends on the size of the profit the firm was able to obtain from the market
 - $\Rightarrow$ this in turn depends on how steep the demand curves are = **elasticities of demand**

Elasticities of Demand

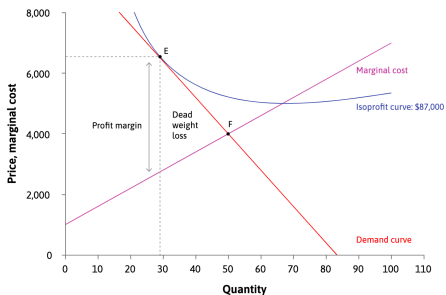
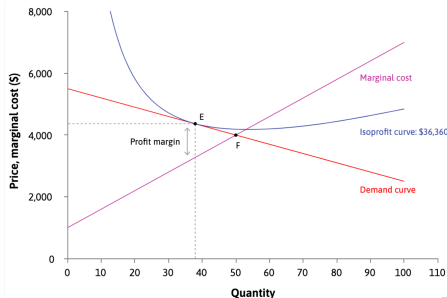
- the (price) elasticity of demand is the degree of responsiveness of consumers to a price change = % change in demand due to a 1% increase in price

$$\epsilon = \frac{-\% \text{ change in demand}}{\% \text{ change in price}} = \frac{\Delta Q/Q}{\Delta P/P}$$

- this is a function of the slope of the demand function
- demand is **elastic** if $\epsilon > 1$ = responsive and **inelastic** if $\epsilon < 1$ = less responsive
- this is important because profit margins for a firm depends not just on the slope (impact of change in price on change in quantity) but also on the elasticity (how many units of goods at the current price level would such a price change apply to compared to at another price level)
 - few available substitutes = hard for consumers to move away = relative inelastic demand = larger price-setting power for firms = higher profits (mark-up) for firms

Elasticities of Demand

- on the left, when demand is elastic (i.e. very responsive), then any change in price will lead to a large change in quantity, so any attempt to increase profits will lead to a large fall in quantity and less profit $\Rightarrow$ so smaller profit margin
- on the right, when demand is inelastic (i.e. less responsive), then any change in price will lead to a small change in quantity, so an increase in the profit margin will increase profits since quantities don't change as much $\Rightarrow$ so larger profit margin



Problem Set 5: Exercise 2 - Cost Function and Profit Maximisation

Consider a company that produces toys. This company has fixed production costs of 100 euros related to the rental of a machine. Moreover, we assume the marginal cost of producing a toy to be constant at 10 euros.

- 1 Draw the cost function (= total cost function) in this case.
- 2 What do the marginal and average cost functions look like ?

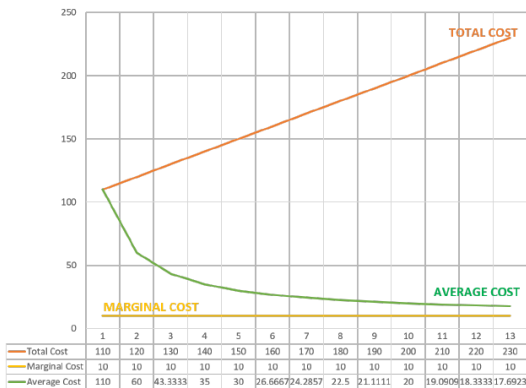
Now suppose that the marginal cost of producing a toy is increasing.

- 3 Explain why the marginal cost might be increasing. Then draw total cost, marginal cost, and average cost curves for this case.
- 4 In another graph, again draw the marginal cost curve and two isoprofit curves.
- 5 Next, draw a demand curve in this graph. (For simplicity, assume that the demand curve is a straight line.) At what point on the graph is the firm's profit maximized ?
- 6 What can be said about the marginal revenue and marginal cost at this point ?

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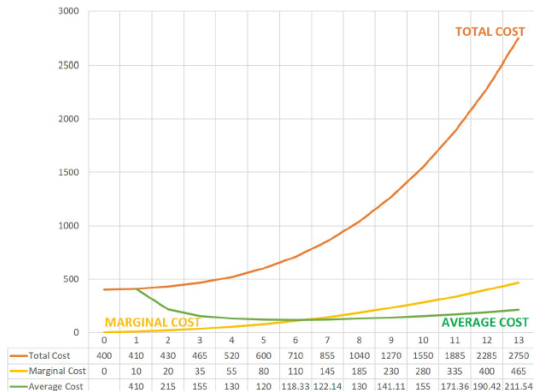
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Problem Set 5: Exercise 2 - Comments

Now suppose that the marginal cost of producing a toy is increasing.

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- things to notice: MC crosses AC at the minimum

Problem Set 5: Exercise 2 - Comments

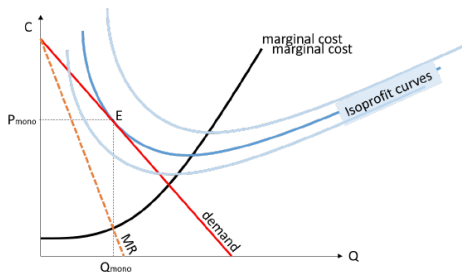
Now suppose that the marginal cost of producing a toy is increasing.

- Explain why the marginal cost might be increasing. Then draw total cost, marginal cost, and average cost curves for this case.
- marginal cost can be increasing (and convex) given diminishing marginal productivity of factors of production
 - recall that if there is diminishing MPK or MPL, then the first marginal unit of K or L produces a lot, then each subsequent units produces less and less
 - say the first unit produces 4 toys: then each toy needed just $1/4$ units of K
 - if the second unit produces 2 toys: then now each toy needs $1/2$ unit of K
 - if the third unit produces 1 toy: then each toy needs 1 unit, etc.
- can also be simply that when we produce near our "production frontier", any additional marginal unit of factor of production is just rarer and thus more expensive (think about lithium, rare metals, etc.)

Problem Set 5: Exercise 2 - Comments

Now suppose that the marginal cost of producing a toy is increasing.

- ④ In another graph, again draw the MC curve and two isoprofit curves.
- ⑤ Next, draw a demand curve in this graph. At what point on the graph is the firm's profit maximized ?
- ⑥ What can be said about the marginal revenue and marginal cost at this point ?



- firm chooses E (point where isoprofit curve is tangent to demand curve) to determine price and quantity to set
- the quantity Q_{mono} is also the quantity at which $MR = MC$

Extra Exercise

Below is a monopolist producing lumber in an economy.

quantity demanded, q	1	2	3	4	5	6	7
willingness to pay, p	14	12	10	8	6	4	2

quantity supplied, q	1	2	3	4	5	6	7
marginal cost	4	5	6	7	8	9	10

- 1 Calculate total and marginal revenue at each q .
- 2 Plot the demand, MC and MR curves. Show consumer and firm surplus as well as the DWL.
- 3 What is the optimal q and p ?

Next...

- Good luck with the Midterm!
- **PS6 Exercise 2** due in class.
- **Quiz 3** next class, Chapters 3-4.